

Nonquadraticity of Lambert Series Values

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For every integer $q \geq 2$, the Lambert value $F(q) = \sum_{n \geq 1} (q^n - 1)^{-1}$ satisfies no nonzero polynomial of degree at most two over $\mathbb{Q}$. In particular, the Erdős–Borwein constant $F(2)$ is neither rational nor quadratic irrational. More generally, for real algebraic $q > 1$, we exclude every nonzero polynomial relation of degree at most e over $\mathbb{Q}(q)$ whenever $e\lambda \log M(q) < \log q$, where $M(q)$ is the Mahler measure of the base and $\lambda = (42963 + 428220/\pi^2)/202140$. This gives nonquadraticity over $\mathbb{Q}(q)$ for Pisot and Salem bases, $F(\sqrt{m}) \notin \mathbb{Q}(\sqrt{m})$ for every integer $m \geq 2$, and irrationality for rational bases $q = a/b > 1$ with $\log b/\log a < 0.5728$. The proof combines positive moment determinants, local denominator bounds, and the product formula.

1 Introduction

The Lambert series

$$F(q) = \sum_{n=1}^{\infty} \frac{1}{q^n - 1}, \quad q > 1,$$

converges absolutely. Erdős proved that $F(q)$ is irrational at every integer base $q \geq 2$ in 1948 [1]. We prove nonquadraticity at every integer base, including the Erdős–Borwein constant $F(2)$.

Theorem 1.1 (Integer bases). *For every integer $q \geq 2$, the numbers $1, F(q), F(q)^2$ are linearly independent over $\mathbb{Q}$.*

We obtain this result from a criterion over $\mathbb{Q}(q)$ for arbitrary real algebraic bases $q > 1$. The criterion compares the degree of a proposed relation with $\log q/\log M(q)$, where $M(q)$ is the Mahler measure of the base. It also applies to Pisot, Salem, and radical bases.

For rational bases $q = a/b > 1$, the same theorem improves the known bounds for Erdős Problem 1049,² enlarging the irrationality region from $\log b/\log a < 0.405683\dots$ to $\log b/\log a < 0.5728$. See Erdős [2, p. 102] for the problem. The earlier bound of Bundschuh–Väänänen [3, Theorem 2] was $1/2 - 1/\pi^2 = 0.398678\dots$; Cook’s 2026 refinement of Zudilin’s permutation method [5, 4] gives the former threshold.

Positive moment determinants provide nonvanishing and decay at the distinguished real embedding. Local denominator bounds produce integral polynomials, while uniform complex estimates control their values at the other conjugates. The product formula then excludes algebraic relations of low degree. The method belongs to the same family as Fauzan’s 2026 preprint on $\zeta(5)$ [6]: positive moment determinants with local denominator control, with cyclotomic arithmetic as the q -specific ingredient here.

¹*AI disclosure:* Generative AI was used extensively in the mathematical investigation, proof development, literature review, exposition, and Lean formalization, under the author’s direction and review.

²<https://www.erdosproblems.com/1049>.

1.1 The general degree criterion

For a real algebraic number q , put $K = \mathbb{Q}(q)$ and let

$$M(q) = |a_g| \prod_{j=1}^g \max(1, |q_j|)$$

be the Mahler measure of its primitive irreducible integer polynomial $a_g \prod_j (X - q_j)$. Define

$$C = 42963 + \frac{428220}{\pi^2}, \quad S = 202140, \quad \lambda = \frac{C}{S} = 0.4271829289 \dots \quad (1)$$

Theorem 1.2. *Let $q > 1$ be real algebraic and $e \geq 0$ an integer. If*

$$e\lambda \log M(q) < \log q, \quad (2)$$

then $P(F(q)) \neq 0$ for every nonzero $P \in K[X]$ of degree at most e .

Proof of Theorem 1.1. For an integer $q \geq 2$, $K = \mathbb{Q}$ and $M(q) = q$. Since $2\lambda < 1$, Theorem 1.2 applies with $e = 2$. $\square$

Corollary 1.3. *If $q > 1$ is an algebraic integer whose other conjugates have modulus at most one, then $1, F(q), F(q)^2$ are linearly independent over $\mathbb{Q}(q)$.*

Proof. Here $M(q) = q$, and $2C < S$. Apply Theorem 1.2 with $e = 2$. $\square$

In particular, this applies to every Pisot or Salem base.

Corollary 1.4 (Rational bases). *Let $a > b > 0$ be integers and let $e \geq 0$ be an integer. If*

$$\frac{\log b}{\log a} < 1 - e\lambda,$$

then no nonzero polynomial in $\mathbb{Q}[X]$ of degree at most e vanishes at $F(a/b)$. In particular, $F(a/b)$ is irrational whenever $\log b / \log a < 0.5728$.

Proof. For coprime a, b , $M(a/b) = a$, so the assertion follows from Theorem 1.2. In general $M(a/b) \leq a$, which gives the same sufficient condition. Finally $1 - \lambda = 0.5728170710 \dots > 0.5728$. $\square$

Corollary 1.5. *Suppose $q > 1$ and $q^n = m$ for positive integers n, m . Put $g = [\mathbb{Q}(q) : \mathbb{Q}]$. If $e \geq 0$ is an integer and $eg\lambda < 1$, no nonzero polynomial over $\mathbb{Q}(q)$ of degree at most e vanishes at $F(q)$. In particular,*

$$F(\sqrt{m}) \notin \mathbb{Q}(\sqrt{m}) \quad (m \in \mathbb{Z}, m \geq 2).$$

Proof. The number q is integral and every conjugate has modulus q , since its n th power is m . Hence $\log M(q) = g \log q$. For the last assertion take $e = 1$ and use $g \leq 2$ and $2\lambda < 1$. $\square$

2 Positive determinants and their normalization

For an indeterminate base U , define a linear functional by

$$\mu_X(t^r) = \frac{1}{U^{r+1} - 1}, \quad \mu_X\left(\frac{1}{U^j - t}\right) = X - S_j(U), \quad S_j(U) = \sum_{v=1}^j \frac{1}{U^v - 1}. \quad (3)$$

Partial fractions extend it to rational functions with distinct poles among the powers of U . For integers $h \geq 1$ and $k \geq 0$, put

$$B_{h,k}(t) = \prod_{j=k}^{k+h-1} (U^j - t), \quad \Delta_{h,k}(U, X) = \det_{0 \leq i, j < h} \mu_X\left(\frac{t^{k+i+j}}{B_{h,k}(t)}\right). \quad (4)$$

Each entry is affine in X , so $\deg_X \Delta_{h,k} \leq h$. The parameter k shifts both the poles and the moments.

For $q > 1$, let $d\nu_q = \sum_{v \geq 1} q^{-v} \delta_{q^{-v}}$. Direct summation gives

$$\int t^r d\nu_q(t) = \frac{1}{q^{r+1} - 1}, \quad \int \frac{d\nu_q(t)}{q^j - t} = F(q) - S_j(q).$$

On its support $0 < t \leq q^{-1}$, every factor $q^j - t$ is positive, so the weight $t^k/B_{h,k}(t)$ is strictly positive. Evaluation of (4) at $(q, F(q))$ is therefore the Gram determinant of $1, t, \dots, t^{h-1}$ for a positive measure with infinite support. The matrix is positive definite.

Define

$$G_{h,k} = k \left(hk + \frac{h(h-1)}{2} \right), \quad Q_{h,k}(U, X) = U^{-G_{h,k}} \Delta_{h,k}(U, X). \quad (5)$$

This normalization makes every coefficient bounded at infinity.

Proposition 2.1. *For $q > 1$,*

$$Q_{h,k}(q, F(q)) > 0, \quad \log Q_{h,k}(q, F(q)) = -\mathcal{T}_{h,k} \log q + O_q(h), \quad (6)$$

where

$$\mathcal{T}_{h,k} = G_{h,k} + h \sum_{j=k}^{k+h-1} j + \frac{h(h+1)(2h+1)}{6} + \frac{kh(h+1)}{2}. \quad (7)$$

For every fixed $c_0 > 0$, $\rho > 1$, and $Z \geq 0$, if $h, k \leq c_0 N$, then uniformly for $|u| = \rho$ and $|z| \leq Z$,

$$|Q_{h,k}(u, z)| \leq \exp\left(O_{\rho, Z, c_0}(N \log(N+1))\right). \quad (8)$$

Every coefficient of $Q_{h,k}$ has degree at infinity at most zero.

Proof. Put $\gamma_q = \prod_{v \geq 1} (1 - q^{-v}) > 0$. On the support of ν_q ,

$$q^{-\sum_{j=k}^{k+h-1} j} \leq B_{h,k}(t)^{-1} \leq \gamma_q^{-1} q^{-\sum_{j=k}^{k+h-1} j}.$$

Comparison of positive Gram forms bounds $\Delta(q, F(q))$ between the corresponding h th powers of these bounds times

$$\det_{0 \leq i, j < h} \frac{1}{q^{k+i+j+1} - 1}.$$

The Cauchy determinant formula factors this expression into its leading power

$$q^{-h(h+1)(2h+1)/6 - kh(h+1)/2}$$

and products of factors $1 - q^{-v}$. Their logarithms contribute $O_q(h)$, uniformly in $k \geq 0$. Multiplication by $q^{-G_{h,k}}$ proves (6).

For the remaining assertions put

$$\mathcal{C}_{i,j}(U, X) = U^{ij}(X - S_j(U)) - \sum_{v=1}^i \frac{U^{j(i-v)}}{U^v - 1}, \quad V_{h,k}(U) = \prod_{0 \leq i < j < h} (U^{k+j} - U^{k+i}).$$

Partial fractions, followed by the change from the simple-pole basis to $1, t, \dots, t^{h-1}$, give

$$\det_{0 \leq i, j < h} \mathcal{C}_{k+i, k+j}(U, X) = (-1)^{\binom{h}{2}} V_{h,k}(U) \Delta_{h,k}(U, X). \quad (9)$$

The entry $\mathcal{C}_{i,j}$ has degree at infinity at most ij . Rearrangement shows that the largest permutation degree in the determinant is $\sum_{i < h} (k+i)^2$. Subtracting $\deg V_{h,k} = \sum_{i < h} i(k+i)$ leaves exactly $G_{h,k}$. This proves the coefficientwise infinity bound.

For $\rho = |u| > 1$,

$$|\mathcal{C}_{i,j}(u, z)| \leq \rho^{ij} \left(|z| + \frac{i+j}{\rho-1} \right), \quad |V_{h,k}(u)| \geq \gamma_\rho^h \rho^{\sum_{i < h} i(k+i)}.$$

Apply the same rearrangement to each permutation term in (9) and cancel $\rho^{G_{h,k}}$. The remaining factorial and entry factors have logarithm $O_{\rho, Z, c_0}(N \log(N+1))$, proving (8). $\square$

3 Local denominator bounds

Identity (9) shows that coefficient denominators are supported at $U = 0$ and the cyclotomic polynomials $\Phi_n(U)$. We bound their orders before specializing the base.

3.1 The zero place

Work in the Laurent series field at $U = 0$, and put

$$P(t) = \prod_{v=k}^{k+h-1} (t - U^v), \quad x_j = U^j.$$

At a pole x_j , write $P_j(t) = P(t)/(t - x_j)$. The finite part of $p(t)/P(t)$ there is

$$\frac{p'(x_j)}{P_j(x_j)} - \frac{p(x_j)P'_j(x_j)}{P_j(x_j)^2}.$$

For an integer $R > k + h$, put

$$Z_R = X + \sum_{v=1}^R \frac{U^v}{1 - U^v}.$$

The functional obtained by summing $-x_j$ times these finite parts for $0 \leq j < R$, and subtracting $Z_R \sum_{k \leq j < k+h} p(x_j)/P_j(x_j)$, converges to $\mu_X(p/P)$. Indeed, polynomial division reduces this statement to

$$-\sum_{j=0}^{R-1} U^{j(r+1)} \longrightarrow \frac{1}{U^{r+1} - 1}$$

and the simple-pole identity obtained by separating $j < v$ and $j > v$ in $\sum_{j < R, j \neq v} U^j / (U^j - U^v)$. The correction Z_R gives the limit $S_v(U) - X$ at the pole $t = U^v$.

Write this finite functional as $\sum_{j < R} \{W_j p(x_j) + \Gamma_j x_j p'(x_j)\}$. Its weights are

$$(W_j, \Gamma_j) = \begin{cases} \left(-\frac{x_j}{P(x_j)}, 0 \right), & j \notin [k, k+h), \\ \left(\frac{x_j P'_j(x_j)/P_j(x_j) - Z_R}{P_j(x_j)}, -\frac{1}{P_j(x_j)} \right), & j \in [k, k+h). \end{cases}$$

Since $\text{ord}(U^j - U^v) = \min(j, v)$ for $j \neq v$, and both Z_R and $x_j P'_j(x_j)/P_j(x_j)$ have nonnegative order, each weight has order at least

$$\omega_j = j - \sum_{v=k}^{k+h-1} \min(j, v).$$

The resulting finite moment matrix has entries $\sum_{j < R} (W_j + (k+i+r)\Gamma_j)x_j^{k+i+r}$. At each pole this is the product of the two-column block $(x_j^i, ix_j^i)^\top$ and the two-row block $((W_j + (k+r)\Gamma_j)x_j^{k+r}, \Gamma_j x_j^{k+r})^\top$. Outside the pole window only the evaluation column remains. The sign change $P = (-1)^h B_{h,k}$ does not affect orders.

Lemma 3.1. *Put*

$$w(j, r) = (k+1)j - \sum_{v=k}^{k+h-1} \min(j, v) + 2(h-1-r)j, \quad 0 \leq r < h.$$

A lower bound for $\text{ord}_{U=0} \Delta$ is any lower bound for $\sum_{r < h} w(j_r, r)$ over admissible selections. It is sufficient to impose the relaxed capacity conditions

$$j_r \geq r \quad (r < k), \quad j_r \geq k + \left\lfloor \frac{r-k}{2} \right\rfloor \quad (r \geq k).$$

The same lower bound holds for every coefficient in X .

Proof. In each finite block, Cauchy–Binet expands the determinant over distinct columns. Before the pole window there is only one slot per node; allowing two slots per node thereafter can only enlarge the set of selections. This gives the displayed inequalities. Order the selected

node indices increasingly as $j_0, \dots, j_{h-1}$. In each of the two minors, every permutation of the exponents $0, \dots, h-1$ has cost at least $\sum_r (h-1-r)j_r$, by rearrangement. The two minors therefore contribute at least twice this sum. The shift contributes $k \sum_r j_r$, and the weight contributes the remaining terms of w .

A sum has valuation at least the minimum valuation of its summands. The bounds therefore hold for every finite block and pass to the Laurent-series limit. To obtain coefficientwise bounds, evaluate X at $h+1$ distinct rational constants, each of valuation zero, and interpolate: the inverse Vandermonde has constant entries and introduces no negative order. This gives the coefficientwise bound. $\square$

3.2 Cyclotomic places

Write $x_+ = \max(x, 0)$ and

$$v_h(n) = \sum_{j \geq 1} (h - jn)_+, \quad \tau_{h,k}(n) = (k + h - \max(k, n))_+.$$

The first quantity is the order of $V_{h,k}(U)$ at Φ_n ; the second counts the indices in $[k, k+h)$ that are at least n .

Lemma 3.2. *Every coefficient of $\Delta_{h,k}$ has cyclotomic pole order at most*

$$h + v_h(n) - \max\{v_h(2n), h - 2\tau_{h,k}(n), 0\}. \quad (10)$$

Proof. At a primitive n th root ζ , use the local coordinate $U = \zeta e^t$. The square matrix

$$\mathcal{M}(t) = \left(t\mathcal{C}_{i,j}(U, X) \right)_{k \leq i, j < k+h}$$

is regular at $t = 0$, and (9) gives $\det \mathcal{M} = \pm t^h V_{h,k} \Delta_{h,k}$. It remains to bound the order of $\det \mathcal{M}$ in two ways.

First,

$$\frac{t\mathcal{C}_{i,j}(U, X)}{U^{ij}} = tX - \sum_{v=1}^j \frac{t}{U^v - 1} - \sum_{v=1}^i \frac{tU^{-jv}}{U^v - 1}.$$

Its constant term is $-(H_{[i/n]} + H_{[j/n]})/n$, where $H_m = \sum_{v=1}^m 1/v$. For each $\ell \geq 1$, the coefficient of t^ℓ in $tU^{-jv}/(U^v - 1)$ is, on each residue class of v modulo n , a polynomial in v of degree at most $\ell - 1$. For $n \mid v$, expand $v^{-1}(vt/(e^{vt} - 1))e^{-jvt}$; otherwise the denominator has nonzero constant term. Summation over $1 \leq v \leq i$ raises the degree by at most one. Thus, after adding the row harmonic term, each coefficient of order ℓ is polynomial in i of degree at most ℓ on each residue class. Multiplication by U^{ij} preserves this bound by convolution.

Consequently the coefficient vectors of order ℓ of every column of $\mathcal{M}$ lie in the span of

$$i^a \mathbf{1}_{i \equiv v \pmod{n}}, \quad H_{[i/n]} i^a \mathbf{1}_{i \equiv v \pmod{n}}, \quad 0 \leq a \leq \ell, \quad 0 \leq v < n.$$

These nested spaces have dimension at most $2n(\ell + 1)$. In a nonzero term of the determinant's Taylor expansion, at most $2n\ell$ chosen coefficient vectors can have order below ℓ . Its total order is therefore at least $\sum_{\ell \geq 1} (h - 2n\ell)_+ = v_h(2n)$.

Second, the constant matrix has entries

$$\mathcal{M}(0)_{i,j} = -\frac{\zeta^{ij}}{n} \left(H_{\lfloor i/n \rfloor} + H_{\lfloor j/n \rfloor} \right).$$

The first summand is supported on $\tau_{h,k}(n)$ rows and the second on that many columns. Its rank is at most $2\tau_{h,k}(n)$, so its corank is at least $(h - 2\tau_{h,k}(n))_+$. Constant invertible row operations make that many rows divisible by t , giving the same lower bound on $\text{ord det } \mathcal{M}$. Combining the two bounds and subtracting $h + v_h(n)$ proves the lemma. All operations retain X as a polynomial parameter, so the assertion holds coefficientwise. $\square$

3.3 Integer descent

By (9), every coefficient of $Q_{h,k}$ lies in

$$\mathcal{R} = \mathbb{Z}[U^{\pm 1}, (U^m - 1)^{-1} : m \geq 1],$$

since the determinant identity divides only by powers of U and differences of powers. Let $J(U)$ be a monic integer polynomial such that every coefficient of $JQ_{h,k}$ has nonnegative order at $U = 0$ and at each cyclotomic factor. These are all possible finite poles, so $JQ_{h,k} \in \mathbb{Q}[U, X]$. For a coefficient of $Q_{h,k}$ written as $A(U)/B(U)$ with $A \in \mathbb{Z}[U]$ and $B \in \mathbb{Z}[U]$ monic, polynomiality says that B divides JA over $\mathbb{Q}[U]$. Gauss's lemma then gives the quotient in $\mathbb{Z}[U]$. Hence

$$\mathcal{P}(U, X) = J(U)Q_{h,k}(U, X) \in \mathbb{Z}[U, X], \quad \deg_U \mathcal{P} \leq \deg J, \quad \deg_X \mathcal{P} \leq h. \quad (11)$$

The first degree bound follows from Proposition 2.1.

4 The arithmetic and analytic scales

Take $h = 60N$ and $k = 3N$, with $N \geq 1$, and write $Q_N = Q_{60N, 3N}$. Equations (5) and (7) give

$$G_N = 5940N^3 - 90N^2, \quad \mathcal{T}_{60N, 3N}/N^3 \longrightarrow S.$$

We clear the local denominators by

$$J_N(U) = U^{a_N} \prod_{n \geq 1} \Phi_n(U)^{c_N(n)}.$$

For each slot r , the cost in Lemma 3.1 is minimized under its relaxed lower bound on j_r by

$$j_r = \begin{cases} r, & r < 3N, \\ \max\{\lfloor (r + 3N)/2 \rfloor, 2r - 60N\}, & 3N \leq r < 60N. \end{cases}$$

Inside the pole window, twice the cost is $j(j + 120N - 1 - 4r) + (3N)(3N - 1)$. The displayed choice minimizes this integral quadratic subject to the slot bound; the exterior affine pieces

give no smaller value. Minimizing separately in each slot gives a lower bound for every admissible selection. The three ranges are $0 \leq r < 3N$, $3N \leq r < 41N$, and $41N \leq r < 60N$. In the middle range write $r = 3N + 2t$ or $3N + 2t + 1$, so that $j_r = 3N + t$ for $0 \leq t < 19N$. In the last range $j_r = 2r - 60N$. The standard sums of integers and squares give

$$\begin{aligned} \sum_{r=0}^{3N-1} w(j_r, r) &= \frac{531}{2}N^3 - 90N^2 + \frac{1}{2}N, \\ \sum_{r=3N}^{41N-1} w(j_r, r) &= \frac{20216}{3}N^3 - 361N^2 + \frac{19}{3}N, \\ \sum_{r=41N}^{60N-1} w(j_r, r) &= -\frac{102163}{6}N^3 + 361N^2 + \frac{19}{6}N. \end{aligned}$$

Adding these expressions gives

$$-\sum_{r < 60N} w(j_r, r) = 10023N^3 + 90N^2 - 10N.$$

Adding G_N yields the clearing exponent

$$a_N = 15963N^3 - 10N. \quad (12)$$

For cyclotomic places with $2n \leq h$, use the filtration term of Lemma 3.2, giving $h + v_h(n) - v_h(2n)$. For $2n > h$, we have $v_h(n) = (h - n)_+$, $v_h(2n) = 0$, and $\tau_{h,k}(n) = (63N - n)_+$, since $n > 30N > k$. The same lemma now gives

$$(h - n)_+ + \min\{h, 2(h + k - n)_+\}.$$

These two bounds are nonnegative and vanish for $n \geq 63N$. To combine them, put

$$E_h(n) = 3(h - n)_+ - (h - 2n)_+ + \sum_{j \geq 1} (h - (2j + 1)n)_+.$$

For $2n \leq h$, separating the odd terms of $v_h(n)$ shows that $E_h(n) = h + v_h(n) - v_h(2n)$. For $2n > h$, it reduces to $3(h - n)_+$. The identity $\min(h, 2x_+) = 2x_+ - 2(x - h/2)_+$ therefore gives the common formula

$$\begin{aligned} c_N(n) &= E_{60N}(n) + 2(63N - n)_+ - 2(33N - n)_+ \\ &\quad - 2(60N - n)_+ + 2(30N - n)_+. \end{aligned} \quad (13)$$

Let $d_N = \deg J_N = a_N + \sum_n \varphi(n)c_N(n)$ and $u_N = \sum_n c_N(n)$. The summatory totient estimate $\sum_{n \leq x} \varphi(n) \sim 3x^2/\pi^2$ [8, 27.11.6], by partial summation, gives

$$\frac{1}{N^3} \sum_n \varphi(n)(KN - mn)_+ \longrightarrow \frac{K^3}{\pi^2 m^2}.$$

The sums are bounded uniformly by a constant times m^{-2} , so this limit may be summed over m . Using $\sum_{j \geq 1} (2j + 1)^{-2} = \pi^2/8 - 1$, we obtain

$$\frac{1}{N^3} \sum_n \varphi(n)E_{60N}(n) \longrightarrow 27000 + \frac{378000}{\pi^2}.$$

The four translated tents contribute $2(63^3 - 33^3 - 60^3 + 30^3)/\pi^2 = 50220/\pi^2$. Together with (12), this proves

$$\frac{d_N}{N^3} \longrightarrow 15963 + 27000 + \frac{378000 + 50220}{\pi^2} = C, \quad u_N = o(N^3). \quad (14)$$

For the last assertion, use $c_N(n) \leq 60N + v_{60N}(n)$ on $n < 63N$ and $v_{60N}(n) \leq (60N)^2/(2n)$, giving $u_N = O(N^2 \log(N+1))$.

By (11), $\mathcal{P}_N(U, X) = J_N(U)Q_N(U, X) \in \mathbb{Z}[U, X]$, with degrees at most d_N in U and $60N$ in X . For fixed $q > 1$, Möbius inversion gives $\log \Phi_n(q) = \varphi(n) \log q + O_q(1)$, uniformly in n . Consequently $\log J_N(q) = d_N \log q + O_q(u_N)$, and Proposition 2.1 gives

$$\mathcal{P}_N(q, F(q)) > 0, \quad \frac{\log \mathcal{P}_N(q, F(q))}{N^3} \longrightarrow (C - S) \log q. \quad (15)$$

5 The product-formula argument

Lemma 5.1. *For fixed $u \in \mathbb{C}$, $Z \geq 0$, and $\epsilon > 0$, eventually, uniformly for $|z| \leq Z$,*

$$|\mathcal{P}_N(u, z)| \leq \exp((C \log \max(1, |u|) + \epsilon)N^3).$$

Proof. For $\rho > 1$, Möbius inversion and the power-series expansion of $\log(1-w)$ give

$$|\log |\Phi_n(u)| - \varphi(n) \log \rho| \leq c_\rho \quad (|u| = \rho),$$

with c_ρ independent of n and the argument of u . Proposition 2.1 and (14) imply the bound $\exp((C \log \rho + o(1))N^3)$ on that circle. Maximum modulus extends it to $|u| \leq \rho$. Given u and ϵ , choose $\rho > \max(1, |u|)$ sufficiently close to that maximum, then take N large. Only the cleared polynomial is evaluated inside the disk, so zero and roots of unity cause no singularity. $\square$

For a number field E , use the unnormalized logarithmic height

$$H_E(x) = \sum_{\sigma: E \hookrightarrow \mathbb{C}} \log \max(1, |\sigma(x)|) + \sum_{v \text{ finite}} \log \max(1, |x|_v),$$

where finite-place absolute values include the local degrees, so the product formula has no additional weights. Complex embeddings are counted individually.

Proposition 5.2. *If a real number field E contains both $q > 1$ and $F(q)$, then*

$$S \log q \leq CH_E(q).$$

Proof. Put $\alpha = F(q)$ and $x_N = \mathcal{P}_N(q, \alpha) \neq 0$. At each finite place, integrality of the polynomial coefficients gives

$$\log |x_N|_v \leq d_N \log \max(1, |q|_v) + 60N \log \max(1, |\alpha|_v).$$

The total α contribution is $O(N)$. At the distinguished real embedding use (15); at every other embedding use Lemma 5.1. Sum the estimates, divide by N^3 , and apply the product formula. Since the number of embeddings is fixed, their ϵ errors can be sent to zero, giving $0 \leq CH_E(q) - S \log q$. $\square$

Proof of Theorem 1.2. The case $e = 0$ is immediate. If a nonzero polynomial of degree at most $e \geq 1$ over $K = \mathbb{Q}(q)$ vanished at $\alpha = F(q)$, then $E = K(\alpha)$ would have relative degree $r \leq e$. The height–Mahler identity [7, Proposition 1.6.6] and the tower formula give

$$H_E(q) = r \log M(q).$$

Proposition 5.2 would imply $S \log q \leq rC \log M(q)$, contradicting (2). For nonintegral bases, the finite-place contribution is accounted for by the leading coefficient in $M(q)$. $\square$

Remark 5.3. Transporting the local bounds through six exact bordered presentations of the same determinant improves the rational-base irrationality threshold to 0.573290; this refinement is formalized in the accompanying Lean development. Numerical parameter searches suggest that this family of estimates saturates near that value.

References

- [1] P. Erdős, *On arithmetical properties of Lambert series*, Journal of the Indian Mathematical Society (N.S.) **12** (1948), 63–66. https://users.renyi.hu/~p_erdos/1948-04.pdf.
- [2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), New Advances in Transcendence Theory, Cambridge University Press, Cambridge, 1988, pp. 102–109. <https://doi.org/10.1017/CB09780511897184.009>.
- [3] P. Bundschuh and K. Väänänen, *Arithmetical investigations of a certain infinite product*, Compositio Mathematica **91** (1994), no. 2, 175–199. https://www.numdam.org/item/CM_1994__91_2_175_0/.
- [4] W. Zudilin, *Heine’s basic transform and a permutation group for q -harmonic series*, Acta Arithmetica **111** (2004), no. 2, 153–164. <https://doi.org/10.4064/aa111-2-4>.
- [5] W. Cook, *Geometric Moments and Rational Lambert Values: Proofs and Further Results*, online manuscript, September 2026. Manuscript and source, accessed 7 October 2026.
- [6] A. Fauzan, $\zeta(5)$ is irrational, preprint, 17 September 2026. <https://doi.org/10.5281/zenodo.22826419>.
- [7] E. Bombieri and W. Gubler, *Heights in Diophantine Geometry*, New Mathematical Monographs **4**, Cambridge University Press, 2006, Proposition 1.6.6. <https://doi.org/10.1017/CB09780511542879>.
- [8] NIST Digital Library of Mathematical Functions, §27.11, equation 27.11.6. <https://dlmf.nist.gov/27.11.E6>.